

MSC CoSim battlecard

Summary/Elevator pitch:

MSC CoSim is a platform that allows you to couple different software tools of the HxGN suite. With MSC CoSim, you can tackle problems requiring coupling of different physical disciplines. It allows a combination of 2 or 3 solvers together while taking care of the solver stepping, orchestrating solving times, triggering mapping of variables etc. With MSC CoSim you have access to a powerful technology that enables truly Multiphysics simulations.

Benefits and USPs

Market positioning

- With MSC CoSim, users can simulate a great variety of applications without compromising on realism and accuracy
- MSC CoSim leverages in one platform all Hexagon's best-in-class solving technology: linear and non-linear FEA, CFD, DEM and MBD
- MSC CoSim allows each discipline expert to work on their tool of competence, (e.g. without forcing the structural engineer to learn how to use a CFD tool)

Some **exemplary applications**:

- Aerospace**: Fluid-Structure interaction, aeroelasticity & flutter of wings/panels
- Marine & Renewables**: stability of floating structures (e.g. offshore platforms, wind turbines, floating solar panels)
- Automotive**: car suspension dynamics, disk brake cooling, airbags, door slamming, fuel tank sloshing
- Industrial machinery**: soft soil (DEM-Adams-FEA), granular mechanics, robotic arms dynamics with deformation etc

Discovery Questions

Identifying challenges

- How do you account for multi-disciplinary phenomena in your design or analyses? Which simulations you couldn't perform due to complexity and computing time?
- How difficult is it to include in your simulations physical disciplines and software you are not expert in? Is that a blocking issue for your daily professional life?
- Global and aggressive competition today means reliability of products is paramount: how do you make sure your simulations are realistic and reliable?
- How many physical prototypes do you create? How much insight do you get from your tests? How time-consuming, inconclusive and costly that can be? Wouldn't a realistic virtual lab be a valuable complementary tool?

May also use/Cross-sell

- MSC CoSim
- MSC Nastran & Marc
- Adams
- Cradle CFD – scFLOW & scSTREAM
- MSC One - for easy access to many tools

Target Industries and Applications

- Auto & transportation, Aerospace & Defense, Marine Engineering, Industrial Machinery, Process Industries, Electronics, Manufacturing

Key Customer Wins/References

- Mitsubishi Heavy Industries, Hitachi R&D, CFD Marine, DSO Singapore, Diehl, Toyota Systems, Airbus, Honda R&D, Japan Steel Works

Pains/Frustrations

- Reliability of single-physics simulations
- Design quality

- Siloed R&D organization with fragmented expertise
- Experts anchored to closed jargon and skills

Solution/Product Positioning

- Co-simulation capabilities leveraging strengths of best-in-class and well-established tools from HxGN
- Coupled simulation technology when timescales of phenomena are similar; robustness of numerical schemes (e.g. floating structures in water, special treatment for high density liquids)

- Co-simulation platform allows seamless communication btw different disciplines / experts

General Objection Handling

Identifying challenges

I mostly need single-physics, only rarely Multiphysics & co-simulations

Suggest to keep using their single-physics tools for their bread-and-butter (e.g. Adams, Cradle, etc) but point out that single-physics can be readily generalized to Multiphysics, when needed, using MSC CoSim

Co-simulations and Multiphysics are complex

Acknowledge. Point out that in cutting-edge R&D you need to get out of your comfort zone. MSC CoSim was in fact developed to alleviate exactly some of these burdens. Usability is improved by each new version of MSC CoSim (released twice per year)

Co-simulations are expensive, as they require purchasing different software

Acknowledge. Emphasize though that investing in a "virtual lab" would reduce the bills of lab testing and prototyping, accelerate time to market and prevent restrictive "tunnel vision" of single-physics approaches. Moreover, MSC One simplifies licensing.

I don't have enough computational power to run CPU-intensive co-simulations

Remind that Cloud and HPC are quickly evolving, allowing for elastic demand and faster computations. Mention our cloud solution Nexus Compute and our in-house AI/ML technology (Odyssee) that in can replace part of the bottlenecks of MSC CoSim with ROMs

Internal use only

To beat the competition, focus on: (1) Identifying existing users who want to go beyond single-physics and propose MSC CoSim; (2) pitching MSC CoSim to prospects interested in Multiphysics, emphasizing AI/ML/ROM, inter-operability, Apex, system-of-systems, and HxGN's best in class single-physics tools; (3) flexibility of licensing scheme (MSC One); (4) highly skilled and helpful support team (e.g. big competitors tend to support well only high-grade customers and frequent buyers); (5) fast pace of software development and expanding capabilities of HxGN

Competitor	Competitor strengths	Objection handling	Key differentiators
Ansys Fluent & CFX	Well-known company and offerings Well established globally with numerous users	Tip: Ansys is the market leader for historical reasons, however their multi-disciplinary solution tends to be fragmented when compared with MSC CoSim's offering. Multiphysics demonstrated mostly on simple, academic examples.	• Expensive • Aging technology • Less practical (overwhelming offering and hard-to-learn tools) • Multiphysics and multidisciplinary are advertised but not really delivered in practice • See recent win at MHI: <i>"If we didn't have MSC CoSim to couple CFD and Nastran, MHI would not have chosen scFLOW, because Fluent is the standard CFD software at MHI"</i>
	Broad solution set	Tip: Broad solution but poorly coupled. MSC CoSim on the contrary couples 2 FEA tools, 2 CFD tools and a MBD tool (with possibility to extend to 3 rd party tools). MSC CoSim capabilities are rapidly expanding. MSC CoSim offers sophisticated and mature techniques like sub-stepping, cutback, virtual mass, initial static analyses etc.	
	Fluent has compliance certification in some industries (eg Nuclear)	Tip: Check the application needed by your prospect. Compliance certification is needed only for some very specific applications.	
	Workbench is well known for connecting ANSYS products	Tip: Workbench is a simulation management tool used to connect Ansys products, e.g. to share parameter values across different tools and to create DoE. It's mostly for pipelined simulations. It does not include all products. ANSYS System Coupling allows co-simulations of CFD with Mechanical/FEA, however ANSYS' investment in this tool is rather sluggish. System Coupling is technically meant only for implicit coupling.	
Siemens STAR-CCM+, Simcenter3D etc.	Deep, senior relationship, particularly with larger customers	Tip: Few large customers get all the attention and soft power: they get the best support; only their requests are incorporated in roadmaps; pharaonic and siloed Simcenter organization currently slows down development	• "FloEFD eats compute resource" • Less practical (overwhelming offering and hard-to-learn tools such as Simcenter 3D) • Tough commercial agreements that can be expensive • No MSC One equivalent • Confusing offering due to acquisition of multiple overlapping products
	Broad offerings across disciplines (e.g. mechanical and electrical design)	Tip: Simcenter has focus on Multidisciplinarity rather than coupled Multiphysics. Very rarely do their different solvers speak to each other. Some co-simulations are technically possible (e.g. STAR-CCM+ with GT, WAVE, Amesim, Abaqus, Simcenter Nastran) however usability is extremely poor.	
	STAR-CCM+ has compliance certification in some industries (eg Nuclear)	Tip: Check the application needed by your prospect. Compliance certification is needed only for some very specific applications.	
	Large sales & support organization	Tip: Sales organization is large and skilled but overpromising and overhyping capabilities, <i>especially</i> around Multiphysics. Many Multiphysics solutions are just academic PoC that fail when used in production	